

SIMATS ENGINEERING

ASSESSMENT TOOL 1

Experiments

NAME:M.Mahammad Rafi
REG NO: 192421364
COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA0716

COURSE OUTCOME COVERED

CO3: *Analyze the performance of core network protocols such as TCP, UDP, HTTP, and DNS under varying conditions*

Assessment Tool Weightage: 40%

SOFTWARE: CISCO PACKET TRACER, WIRESHARK

EXPERIMENTS

QUESTIONS:4

Total Marks:40

1. Design a simple LAN consisting of two PCs connected through a switch using Cisco Packet Tracer. Configure appropriate IP addresses for both hosts and verify connectivity using the ping command. Simultaneously capture the packet exchange using Wireshark. Identify the Ethernet frame fields such as Source MAC Address, Destination MAC Address, EtherType, and Frame Check Sequence (FCS). Explain how the Data Link layer uses MAC addressing to ensure successful frame delivery within the local network.
2. Create a network topology consisting of two LANs connected through a router in Cisco Packet Tracer. Configure IPv4 addressing, subnet masks, and default gateways for all devices, and verify end-to-end communication using the ping command. Capture the ICMP packets using Wireshark and analyze the IPv4 header fields, including Source IP Address, Destination IP Address, Time-To-Live (TTL), Protocol Number, and Header Length. Explain the role of the Network layer in forwarding packets between different networks.
3. Construct a local area network with multiple hosts connected through a switch in Cisco Packet Tracer. Clear the ARP cache on one of the hosts and initiate communication with another host using the ping command. Capture the communication in Wireshark and identify the ARP Request and ARP Reply packets. Analyze the sender and target MAC/IP addresses and explain how the Address Resolution Protocol resolves logical IP addresses into physical MAC addresses before data transmission.
4. Develop a simple network consisting of two PCs connected through a router in Cisco Packet Tracer. Configure the required IP addresses and test connectivity using the ping command. Capture the ICMP traffic using Wireshark and identify the Echo Request and Echo Reply messages. Examine important ICMP fields such as Type, Code, Identifier, and Sequence Number, and calculate the Round Trip Time (RTT). Discuss how ICMP assists in network troubleshooting and connectivity verification.

5. Create a client-server network in Cisco Packet Tracer by configuring a web server and a client PC. Enable the HTTP service on the server and access the hosted webpage from the client using a web browser. Capture the communication in Wireshark and identify the TCP three-way handshake, followed by the HTTP GET request and HTTP response messages. Analyze the application-layer communication and explain how TCP provides reliable delivery for HTTP traffic through connection establishment, acknowledgments, and ordered data transfer.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation

1. Simple LAN with Two PCs Connected Through a Switch

Aim

To design a simple LAN using Cisco Packet Tracer, configure IP addresses, verify connectivity using the ping command, capture packets using Wireshark, and analyze Ethernet frame fields.

Network Topology

PC0 ----- Switch -----PC1

IP Configuration

Device	IP Address	Subnet Mask
PC0	192.168.1.1	255.255.255.0
PC1	192.168.1.2	255.255.255.0

Steps

1.

Open Cisco Packet Tracer.
2.

Place one switch and two PCs.
3.

Connect both PCs to the switch using Copper Straight-Through cable.
4.

Configure IP addresses on both PCs.
5.

Open Command Prompt on PC0 and type:
ping 192.168.1.2
6.

Open Wireshark and start packet capture.
7.

Observe Ethernet frames.

Ethernet Frame Fields

Field	Example	Purpose
Destination MAC	00:11:22:33:44:55	Receiver's MAC address
Source MAC	AA:BB:CC:DD:EE:FF	Sender's MAC address
EtherType	0x0800	Indicates IPv4 packet
Payload	ICMP Data	Actual transmitted data
FCS	CRC Value	Detects transmission errors

Result

- Ping is successful.
- Ethernet frames are captured.

Explanation

The **Data Link Layer (Layer 2)** uses **MAC addresses** to uniquely identify devices within the same LAN. Before transmitting data, the source device places the destination MAC address into the Ethernet frame. The switch forwards the frame only to the intended device based on its MAC address table. The **Frame Check Sequence (FCS)** ensures error detection during transmission.

2. Two LANs Connected Through a Router

Aim

To configure two LANs connected through a router, verify communication, and analyze IPv4 packet headers.

Network Topology

PC0 ---- Switch ---- Router ---- Switch----- PC1

IP Configuration

Device	IP Address	Subnet Mask	Default Gateway
PC0	192.168.1.2	255.255.255.0	192.168.1.1
Router G0/0	192.168.1.1	255.255.255.0	-
Router G0/1	192.168.2.1	255.255.255.0	-
PC1	192.168.2.2	255.255.255.0	192.168.2.1

Router Configuration

```
enable
configure terminal
interface g0/0
ip address 192.168.1.1 255.255.255.0
no shutdown
exit
```

```
interface g0/1
ip address 192.168.2.1 255.255.255.0
no shutdown
end
```

Verification

```
ping 192.168.2.2
```

IPv4 Header Fields

Field	Example	Purpose
Source IP	192.168.1.2	Sender IP
Destination IP	192.168.2.2	Receiver IP
TTL	128	Prevents routing loops
Protocol	1	Indicates ICMP
Header Length 20 Bytes		Size of IPv4 header

Result

- Ping successful between different networks.

Explanation

The **Network Layer (Layer 3)** forwards packets between different networks using IP addresses. Routers examine the destination IP address and forward packets to the correct network while decrementing the TTL value at every hop.

3. ARP Packet Analysis

Aim

To observe ARP Request and ARP Reply packets using Wireshark.

Network Topology

PC0 ----- Switch -----PC1

IP Configuration

Device	IP Address
PC0	192.168.1.10
PC1	192.168.1.20

Procedure

1. Configure both PCs.
2. Clear ARP cache:

```
arp -d
```

3. Ping PC1 from PC0.
4. Capture packets in Wireshark.

ARP Analysis

ARP Request

Field	Value
Sender MAC	PC0 MAC
Sender IP	192.168.1.10
Target MAC	00:00:00:00:00:00
Target IP	192.168.1.20

ARP Reply

Field	Value
Sender MAC	PC1 MAC
Sender IP	192.168.1.20
Target MAC	PC0 MAC
Target IP	192.168.1.10

Result

- ARP Request and Reply successfully captured.

Explanation

ARP resolves a device's **IP address into its physical MAC address** before communication begins. The sender broadcasts an ARP Request, and the destination responds with its MAC address. The sender stores this information in its ARP cache for future communication.

4. ICMP Packet Analysis

Aim

To capture ICMP Echo Request and Echo Reply messages and calculate RTT.

Network Topology

PC0 ----- Router----- PC1

IP Configuration

Device	IP Address
PC0	192.168.1.2
Router	192.168.1.1 / 192.168.2.1
PC1	192.168.2.2

Procedure

1. Configure IP addresses.
2. Ping PC1 from PC0.
3. Capture packets in Wireshark.

ICMP Fields

Field	Example	Purpose
Type	8	Echo Request
Code	0	No Error
Identifier	0x0001	Identifies request
Sequence Number 1		Tracks packet order

Echo Reply:

Field	Value
Type	0
Code 0	

Round Trip Time

RTT = Reply Time – Request Time

Example:

Request = 0.002 s

Reply = 0.005 s

RTT = 3 ms

Result

- ICMP Echo Request and Reply observed.

Explanation

ICMP is used for network diagnostics. The **ping** command sends Echo Requests and receives Echo Replies to verify connectivity and measure network latency (RTT).

5. Client-Server Network (HTTP)

Aim

To configure a web server and client, access a webpage, and analyze TCP and HTTP communication.

Network Topology

Client PC ----- Switch----- Web Server

IP Configuration

Device	IP Address
Client PC	192.168.1.2
Web Server	192.168.1.10

Configuration

1. Configure IP addresses.
2. Enable **HTTP Service** on the server.
3. Open Web Browser on the client.
4. Enter:

http://192.168.1.10

5. Start Wireshark capture.

TCP Three-Way Handshake

Step	Message
1	SYN
2	SYN-ACK
3	ACK

HTTP Communication

Request	Response
HTTP GET HTTP/1.1 200 OK	

TCP Reliability

- **Connection Establishment:** Uses the three-way handshake (SYN, SYN-ACK, ACK).
- **Acknowledgments:** Every received segment is acknowledged to ensure delivery.
- **Ordered Delivery:** Sequence numbers keep data in the correct order.
- **Retransmission:** Lost packets are resent automatically.

Result

- The webpage loads successfully.
- Wireshark captures the TCP handshake, HTTP GET request, and HTTP response.

Explanation

The **Application Layer** uses HTTP to exchange web pages between the client and server. The **Transport Layer** uses TCP to provide reliable communication through connection establishment, acknowledgments, retransmission of lost packets, and ordered delivery of data.